GOOGLE AI BUILDER CUP 2026 · MEDIA, CONTENT & DIGITAL EXPERIENCES

Ember

Turns what your kids watch into what your family does.

Solution & Technical Design Document

Version 1.2 · Pre-build baseline

Theme Media, Content & Digital Experiences
Submission 19 October 2026
Team size 4
Platform Google Cloud Platform (exclusively)
Deployment Cloud Run + Firebase Hosting
Models Gemini via Vertex AI

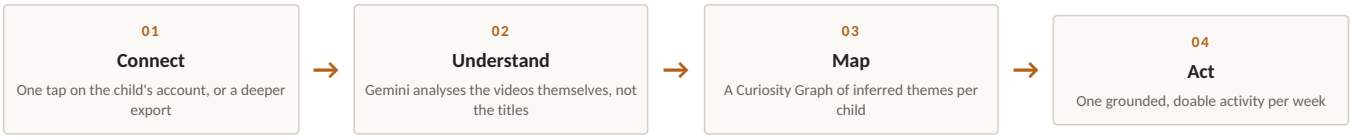
01 Executive Summary

Ember is a family-scoped AI service that reads what a child actually watches, infers what they are genuinely curious about, and turns that curiosity into real-world activity the family does together.

Existing tools in this space measure and restrict. Screen-time dashboards tell a parent *how many hours*; content filters tell a child *no*. Neither answers the question parents actually have, which is what all this watching *means* and what to do with it.

Ember takes a different position. Attention a child has already given something is a signal, not a problem. The product's job is to redirect that attention into a skill, a hobby, or an afternoon with a parent.

The four-step loop



Completion feedback returns to step 3, so the graph sharpens week over week. The loop is the product; any single step is only a feature.

What makes it defensible

Real video understanding	Gemini ingests the actual video via URL and extracts concepts, modality, vocabulary band and latent real-world skill. Competing approaches classify titles and channel names, which cannot distinguish a survival documentary from a survival game stream.
Themes, not tags	The Curiosity Graph surfaces inferred themes with reasoning attached — *"consistently rewatches the segment where something is built from nothing"* — rather than category labels the parent already knew.
Every output ends offscreen	A design constraint, not a feature. You do not solve a screen-time problem by adding a screen. Each generation terminates in a physical, offline action.
Common Ground	Intersecting two siblings' graphs produces an activity neither profile would generate alone. Only possible because the household, not the individual, is the unit of the product.
Household-scoped by design	Data never leaves the family tenant. Sharing happens by generated report, never by granting an outsider access.

> THE QUESTION JUDGES WILL ASK
>
> **"Why doesn't Google build this?"** Because it is YouTube recommending less YouTube. The incentive misalignment is structural. A third party operating on parent-authorised exported data is the natural home for this product, and Google's own Data Portability API is the mechanism that makes it legitimate.

02 Problem Statement

The situation

Children in the 7–12 band consume several hours of short- and long-form video daily. Much of that content is not junk. It includes genuine instruction, craft, science, construction, cooking, survival technique and narrative. But consumption is where it stops. Almost none of it converts into an attempt, a skill, or a shared experience.

What parents can see today, and what they cannot

PARENTS CAN SEE	PARENTS CANNOT SEE	PARENTS ACTUALLY WANT
Total hours watched	What the child is drawn to *underneath* the surface topic	A reason to feel less guilty
Apps used	Whether the content is instructional or purely passive	Something specific to do this weekend
Which channels appear often	What the interest could grow into	A way to connect through the interest rather than fight it
Whether content was age-flagged	Where two children's interests overlap	Evidence their child is developing, not just consuming

Why existing solutions miss

- **Screen-time controls** are subtractive. They cap and block. A cap creates conflict and produces no alternative, so the time is simply displaced onto another screen.
- **Platform recommenders** are optimised for retention. Their definition of a good outcome is another video. They will never propose that the child close the app.
- **Educational apps** ignore the child's existing interests entirely and start from a curriculum. Engagement is therefore something they must manufacture rather than inherit.
- **Parenting-advice content** is generic. It does not know this child.

> PROBLEM STATEMENT, ONE SENTENCE
>
> Children generate an enormous, richly detailed record of what genuinely interests them, and no product turns that record into anything a family can act on.

Theme alignment

The challenge asks for solutions that transform how content is *discovered*, *personalized*, *understood or experienced*. Ember addresses **understanding** (multimodal comprehension of what a child consumes), **personalization** (a per-child inferred interest model), and **experience** (redirecting digital consumption into offline activity). It sits across the media, education and communication ecosystems the theme names explicitly.

03 Users, Scope & Boundaries

The unit is the household, not the bloodline

Ember's tenant is a single household. An aunt with her own children is not a stakeholder in this child's development; she is a relative. Scope is restricted to the adults who actually do the parenting.

ROLE	COUNT	CAPABILITY
Guardian	1–2	Full access to all children in the household. Grants and revokes data consent. Receives insight and reasoning. Approves, edits or rejects proposed activities. Generates shareable reports.
Child	1–4	No login in v1. Receives a printed or guardian-displayed Quest Card. Optionally logs completion via the guardian's device.
Outsider	—	**No account exists.** Teachers, tutors, grandparents and clinicians receive a generated PDF if and only if a guardian chooses to produce and hand one over.

Primary persona

The guardian. Time-poor, mildly guilty about screen time, does not want another dashboard to check, and would act on a specific suggestion but will not construct one from raw data. Success for this user is one decision made for them each week.

Secondary persona

The child, 7–12. Not a user of the software. A recipient of an artifact. Old enough to follow a multi-step challenge, young enough that a parent's involvement is welcome rather than embarrassing.

Explicitly out of scope for v1

- Any child-facing application, feed or account
- Screen-time enforcement, blocking or restriction of any kind
- Platforms other than YouTube
- Children under 6 or over 13
- Cross-household or social features of any description
- Academic assessment, diagnosis, or developmental scoring

> POSITIONING DISCIPLINE
>
> Ember is not a parental control product and must never be described as one. Controls restrict. Ember redirects. Every piece of copy, UI text and pitch language should hold this line, because the moment the product is filed mentally as "another screen-time app" the differentiation evaporates.

04 Solution Overview

4.1 Connect

A guardian onboards the household, adds each child, and chooses how much signal to give Ember. Three tiers are supported, increasing in depth and in friction. All three feed the same downstream pipeline.

TIER	SOURCE	FRICTION	MECHANISM AND SIGNAL QUALITY
0	**Guardian saves a video**	None	A share-sheet target and browser action. The guardian saves anything they come across, exactly as they would on a social platform. **Declared intent, not discovered signal** — the guardian already knows what they saved. Solves cold start and seeds the household on day one, but cannot by itself produce an insight.
1	**Child's liked videos and subscriptions**	One tap	**Default onboarding.** A single OAuth consent on the child's account under `youtube.readonly`. Read `contentDetails.relatedPlaylists.likes` from `channels.list`, then `playlistItems.list` against that playlist; `subscriptions.list` for channels. Real child signal with no export, no email wait and no file handling. Sparser than history — a child may like few videos — but subscription choice is a strong deliberate interest signal, and a channel's recent uploads can be analysed to characterise it.
2	**Full watch history**	High	Takeout JSON upload, or the Data Portability API using `InitiatePortabilityArchive` with resource `myactivity.youtube`. Time-based access permits repeat pulls and the Time Filter requests only the new slice. Richest signal by a wide margin and the product's ceiling, offered as "go deeper" rather than as the front door.

> **WHY THE TIERS EXIST**
>
> Requiring a Takeout export as the first thing a family does would lose most non-technical households before the product ever demonstrated value. Tier 1 removes almost all of that friction while preserving what matters — that the signal comes from *the child*, not from the guardian. Tier 0 alone would invert the product: if the guardian supplies the input, Ember can only reflect back what they already believed, and the Curiosity Graph has nothing to infer.

DEAD END, RECORDED FOR COMPLETENESS

The YouTube Data API cannot return watch history. The `contentDetails.relatedPlaylists.watchHistory` property was deprecated in 2016 and always returns the placeholder value `HL`; no OAuth scope for history exists and the `activities` resource is deprecated. The `favorites` playlist is likewise deprecated. Only `likes`, `uploads` and `subscriptions` remain usable.

THE CONSENT CHAIN

Google's Family Link disclosure for parents states that where a child holds a Google Account, the parent may access, update, remove, **export** and restrict processing of that child's information. Ember relies on precisely this existing right, for every tier. The product never requests access it is not entitled to.

ACCOUNT PREREQUISITE

All child signal attaches to a Google Account, not to a device or an application. Each child must therefore hold their own account. For a child under 13 the guardian creates a supervised Google Account through Family Link; from 13 upward the equivalent is a supervised teen account. In both cases the guardian controls the account and can authorise access on the child's behalf.

OPERATIONAL CONDITIONS

A channel must exist	The likes playlist is only present once the account has a YouTube channel. An account that has never created one returns no `likes` id and tier 1 is unavailable until it does.
History can be disabled	Affects tier 2 only. If YouTube History is paused in the child's activity controls, Google holds nothing to export. Ember needs an onboarding check; the remedy is a settings toggle, not a support ticket.
Shared devices contaminate	Anything watched on a device signed into an account lands in that account — a living-room television deposits the whole family's viewing into whichever account it holds. Principal justification for the confirmation step on screen 5.
YouTube Kids is separate	Viewing inside the YouTube Kids application sits in a Kids profile and its presence in the Takeout history file is unconfirmed. Ember targets supervised YouTube, which is what most 9–12 year olds use, and states that as the requirement.
Exports are not exhaustive	Takeout archives are reported to vary in completeness between runs. Immaterial here, since the product is interested in recent viewing, but the submission should not claim complete history.

Where children share a guardian's account entirely, per-child attribution is not possible and Ember declines rather than guessing. A future release may cluster a commingled history into probable viewer profiles using content pattern and time-of-day signals; this is roadmap, not v1.

OPEN VALIDATION ITEM

Tier 1 has been confirmed against the public API surface but **not yet against a supervised child account specifically**, where Google may restrict API access. The prototype validates the mechanism on a standard test account created to replicate a child's viewing; supervised-account behaviour is an open item. This is stated plainly rather than implied, and tier 2 remains available as the fallback if tier 1 proves unavailable for supervised accounts.

4.2 Understand

The distinguishing technical decision. Ember does not classify titles. It passes the YouTube URL directly to Gemini, which processes the video itself, and extracts a structured record per item.

```
{
  "video_id": "...",
  "concepts":        ["improvised shelter", "load-bearing structure", "knot selection"],
  "modality":        "procedural",        // procedural | narrative | passive | reactive
  "vocabulary_band": "age_9_11",
  "affect":          "calm, methodical",
  "latent_skill":    "building a functional object from found materials",
  "transferable":    true,
  "evidence":        "Sustained demonstration with no cuts during assembly."
}
```

Titles cannot distinguish a survival documentary from a survival game stream, or an instructional baking video from a reaction to one. Video understanding can. This distinction is the foundation everything above it depends on.

4.3 Map — the Curiosity Graph

Per-video records compose into a per-child model of inferred themes, each carrying supporting evidence and a confidence score.

WHAT A TAG-BASED SYSTEM PRODUCES	WHAT THE CURIOSITY GRAPH PRODUCES
Survival · Gaming · DIY *The parent already knew this. No decision follows from it.*	**Improvisation under constraint.** Across 38 videos, Aiden skips the entertainment framing and rewatches the segments where something functional is built from nothing. He has never searched for this. It is an unspoken interest. *A decision follows from this.*

NON-NEGOTIABLE OUTPUT REQUIREMENT

Every theme must carry a named label, a plain-language reasoning statement, and cited evidence from specific videos. A theme without reasoning is a tag, and a screen full of tags is a dashboard. This constraint is enforced in the structured output schema from day one, not corrected by prompt-tuning later.

4.4 Act

Once per week the household receives a single proposed activity. Not a list, not a report to interpret — one thing to do, sized to a weekend, grounded in options that actually exist nearby.

RECIPIENT	RENDERING OF THE SAME GENERATION
Guardian	The activity, the reasoning that produced it, the materials needed, estimated time, and an edit or reject control.
Child	A printable Quest Card. Mission framing, illustrated, no explanation of the analysis behind it. Physical artifact by default.
Second guardian	Logistics only — what, when, what to buy.

Completion is logged with one tap and an optional photo. That signal feeds back into the graph, weighting themes that produced successful activities and down-weighting those that did not.

05 Feature Specification

5.1 Curiosity Graph — primary feature

Purpose	Give a guardian a genuine insight into their child that they did not previously hold.
Input	Structured per-video records from the Content Sense agent.
Output	3–6 named themes, each with reasoning, cited evidence videos, confidence, and a trajectory indicator (emerging, sustained, fading).
Interface	Force-directed graph. Theme nodes sized by weight, video nodes clustered beneath. Selecting a theme reveals the reasoning panel.
Success test	A guardian looking at their own child's graph says something they did not know. If the reaction is "yes, obviously," the feature has failed.

5.2 Common Ground — differentiator

Purpose	Find the theme-level overlap between two siblings and generate an activity that gives each child genuine ownership of part of it.
Mechanism	Semantic intersection of two Curiosity Graphs at theme level, not surface level. Gemini then synthesises a single activity satisfying both, with an explicit statement of which child contributes what.
Quality bar	Surface overlap ("both like dinosaurs") is a coincidence, not a finding. The feature earns its place when the surfaces look unrelated and the themes converge — survival content and baking shorts sharing *making things work with limited ingredients*, resolving to cooking over a campfire.
Why it only exists here	A single-child product cannot produce it. It is a direct consequence of choosing the household as the tenant, and it is the feature that connects the analysis to the family-bonding outcome.

> **CAUSAL LINK TO STATE IN THE PITCH**
>
> Common Ground is only findable *because* the graph is built on inferred themes rather than categories. No tag-level system would ever connect survival content to baking. The two features are causally linked, not bolted together — say so explicitly.

5.3 Quest Card

Purpose	Deliver the activity to the child in a form that moves them away from a screen.
Format	Single-page printable card. Mission title, 3–5 steps, materials, an illustration generated by Imagen on Vertex AI, and a completion box.
Constraint	Physical by default. If a digital fallback is displayed, it is capped at the completion check-in.

5.4 Family Report

Purpose	Let a guardian share insight with an outsider without granting that outsider access.
Model	**Share by artifact, not by access.** A generated PDF is a point-in-time snapshot the guardian chooses to hand over. An account grants ongoing visibility into a child's life; a report does not.
Audience framing	The guardian selects the recipient and Gemini re-frames the same graph accordingly — a teacher receives learning patterns and classroom hooks, a grandparent receives gift ideas and conversation starters, a tutor receives skill gaps. One data source, audience-appropriate generation.
Analogy	A health check report. The guardian owns it, reads it for direction, and decides who else ever sees it.
Controls	Section-level redaction before generation. Footer records the generating guardian and the date. No live link, no revocable share — deliberately a document.

5.5 Guardian Agent — safety layer

Every artifact concerning or shown to a child passes a final gate that checks age-appropriateness, physical safety of the proposed activity, absence of commercial pressure, and tone. Cheap to implement, disproportionately valuable in evaluation, and named explicitly on stage.

5.6 Save to Ember

Purpose	Let a guardian capture something they come across — a camping video, a craft tutorial — with the same gesture they already use on any social platform. A share-sheet target on mobile and a browser action on desktop.
The critical design decision	A save is an **input to** the Curiosity Graph, never a replacement for it. Saving alone would reduce Ember to a bookmarking tool with suggestions attached, because a guardian's saves only reflect what the guardian already believed.
The interaction	On save, Ember responds using the graph: *"This fits Aiden's improvisation theme. It is a stretch for Mira — but add the cooking element and she is in. Make it this weekend's activity?"* The system states which child it suits and why. Five seconds, instantly legible, and it proves the graph is doing real work.
Secondary roles	Solves cold start before any child data is connected. Acts as a steering signal, weighting the kinds of activity the Quest Planner proposes.

06 Technical Architecture

Google Cloud exclusively. No third-party services, no external model providers, no non-Google managed infrastructure.

6.1 Platform layers

PRESENTATION

Firebase Hosting — guardian web app (SPA) | **Firebase Authentication** — guardian identity | **Firebase App Check** — client attestation

APPLICATION & ORCHESTRATION

Cloud Run — single container, API + agent runtime | **Agent Development Kit** — four-agent graph | **Cloud Tasks** — async ingestion fan-out | **Cloud Scheduler** — weekly cycle trigger

INTELLIGENCE

Vertex AI — Gemini Flash video understanding, high volume | **Vertex AI — Gemini Pro** graph synthesis, Common Ground | **Grounding with Google Search** real local options | **Vertex AI Embeddings** theme similarity | **Imagen on Vertex AI** Quest Card illustration

DATA

Firestore — tenanted household data, vector search on themes | **Cloud Storage** — Takeout uploads, generated PDFs, photos | **Secret Manager** — OAuth client credentials

OPERATIONS

Cloud Build + Artifact Registry | **Cloud Logging / Monitoring** | **IAM** least-privilege service accounts

> **DEPLOYMENT DECISION**
>
> Four *agents*, one *service*. The agents are defined as a graph within ADK and deployed as a single Cloud Run container. This preserves the agentic architecture the challenge rewards while leaving one deployment target, one log stream and one failure surface to debug under time pressure.

6.2 Agent design

AGENT	MODEL	RESPONSIBILITY
Content Sense	Gemini Flash	Consumes watch records. Passes each YouTube URL to Gemini for native video understanding. Emits the structured per-video schema. Aggressively cached by video ID, since videos are global and immutable — cache hit rate rises with every household onboarded.
Curiosity Graph	Gemini Pro	Synthesises per-video records into named themes with reasoning, evidence and confidence. Computes embeddings for theme-level comparison. Applies completion feedback to re-weight. Persists to Firestore.
Quest Planner	Gemini Pro + Search grounding	Takes one or two graphs and produces a single proposed activity. **Must be grounded** — an invented local class is a hallucination that will be caught. Owns Common Ground synthesis when two children are present.
Guardian	Gemini Flash	Terminal gate on every child-facing or child-describing artifact. Age-appropriateness, physical safety, tone, commercial neutrality. Blocks or returns for regeneration.

6.3 Request flow

```
INGESTION  (async, minutes)
  Guardian uploads Takeout JSON  ->  Cloud Storage
  Cloud Run parses, dedupes, normalises  ->  Firestore (raw records)
  Cloud Tasks fan-out, one task per uncached video
      -> Content Sense -> Vertex AI Gemini Flash (file_data: YouTube URL)
      -> structured record -> Firestore (shared video cache)
  Completion -> trigger Curiosity Graph agent

SYNTHESIS  (async, ~1 min)
  Curiosity Graph -> Gemini Pro -> themes + reasoning + evidence
                  -> Vertex AI Embeddings -> theme vectors
                  -> Firestore: families/{id}/graph/{childId}

WEEKLY CYCLE  (Cloud Scheduler, Thursday)
  For each household:
      load graphs for all children
      if >1 child: Common Ground intersection via embedding similarity
      Quest Planner -> Gemini Pro + Search grounding -> activity
      Guardian agent -> safety gate
      Imagen -> Quest Card illustration
      render PDF -> Cloud Storage -> notify guardian

FEEDBACK
  Completion tap + optional photo -> Firestore
                                  -> re-weight themes on next synthesis
```

6.4 Latency posture

Processing a month of history is not an interactive operation. The architecture is asynchronous by design from the first commit: upload returns immediately, work proceeds through Cloud Tasks, and the guardian is notified on completion. A considered progress experience is a first-class UI deliverable, not a spinner added late.

07 Data Model & Tenancy

7.1 Firestore structure

```
families/{familyId}
  ├─ meta                 createdAt, region, locale
  ├─ members/{userId}     role: guardian
  │                       displayName, email, joinedAt
  ├─ children/{childId}   displayName, ageBand
  │                       consentGrantedAt, consentGrantedBy, consentScope
  │                       sourceAccountHash        (never the raw address)
  ├─ sources/{sourceId}   provider, lastSyncAt, method: takeout | dpa
  │                       scopeGranted, expiresAt
  ├─ records/{recordId}   childId, videoId, watchedAt
  ├─ graph/{childId}      themes[]  { label, reasoning, evidence[],
  │                                   confidence, trajectory, vector }
  │                       generatedAt, modelVersion
  ├─ quests/{questId}     childIds[], activity, reasoning, materials,
  │                       groundedRefs[], status, completedAt, feedback
  └─ reports/{reportId}   childId, audience, generatedBy, generatedAt,
                          redactions[], storagePath

videoCache/{videoId}      SHARED, NON-TENANTED
                          concepts[], modality, vocabularyBand, affect,
                          latentSkill, transferable, evidence,
                          analysedAt, modelVersion
```

WHY THE VIDEO CACHE SITS OUTSIDE THE TENANT

A video's content is a property of the video, not of the child who watched it. Holding the analysis in a shared, non-tenanted collection contains no personal data whatsoever, and makes the marginal cost of each additional household fall as the corpus grows. This is the core of the unit-economics story in section 11.

7.2 Isolation

Firebase Authentication issues a `familyId` custom claim at sign-in. Firestore security rules reject any read or write where the token claim does not match the document's family path. Consent records are guardian-write-only. There is no cross-family query path anywhere in the API surface, by construction rather than by filter.

```
match /families/{familyId}/{document=**} {
  allow read, write: if request.auth != null
                     && request.auth.token.familyId == familyId;
}
match /families/{familyId}/children/{childId} {
  allow write: if request.auth.token.familyId == familyId
               && request.auth.token.role == 'guardian';
}
```

SCOPE DISCIPLINE

Technical Merit is weighted at 40% and is scored on meaningful use of generative AI. Access control is table stakes; no hackathon is won on role-based access control. Model the tenancy correctly in the schema, implement the minimum that makes the demonstration honest, and direct remaining hours to the video-understanding layer.

08 Privacy, Consent & Child Safety

Ember profiles the behaviour of children. This must be raised by the team before it is raised by an evaluator.

Principles

Derived rights only	Ember requests no access a guardian does not already hold. The Family Link parent disclosure establishes the guardian's right to export their child's Google data; Ember exercises that existing right with explicit authorisation.
Household boundary	No cross-household features, no social layer, no aggregate benchmarking against other children. There is no code path by which one family's data reaches another.
Share by artifact	Outsiders never receive accounts. They receive documents, produced deliberately, by a guardian, with section-level redaction available before generation.
Minimise at rest	Raw watch records are retained only until the graph is synthesised, then reduced to derived themes. Video analysis is depersonalised and stored in the shared cache. Child email addresses are hashed, never stored raw.
No behavioural advertising	No advertising identifiers, no data sale, no third-party analytics on child data. Ever, and stated in product.
Revocation is real	A guardian withdrawing consent for a child triggers deletion of that child's records, graph, quests and reports, not a flag.
Safety gate	The Guardian agent is a hard dependency on every child-facing artifact, not an optional filter.

What Ember deliberately does not do

- No developmental scoring, ranking or diagnosis of any child
- No comparison of a child against peers or norms
- No surveillance framing — the guardian sees themes and reasoning, never a raw feed of everything their child watched
- No covert operation — the consent record names the guardian who granted it and when

> **COMPLIANCE POSTURE FOR THE PROTOTYPE**
>
> The Data Portability API's YouTube scopes are classified as sensitive and availability is limited to a published list of supported countries. The prototype therefore operates in OAuth testing mode against team-owned accounts, with Takeout upload as the universally available path. This is stated plainly in the submission rather than glossed — a prototype that is honest about its approval status is more credible than one that implies production clearance.

09 Interface & Experience Flow

9.1 Where Ember lives

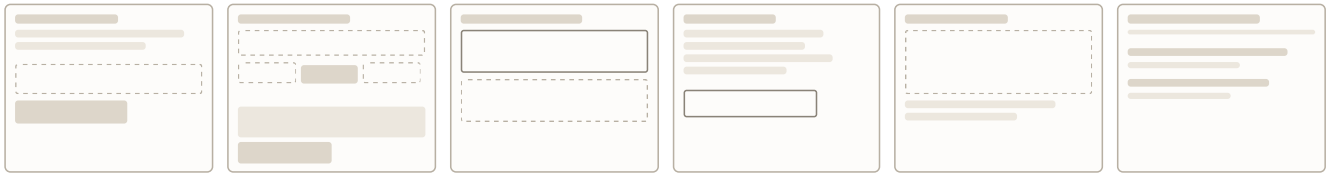
A separate, responsive web application. Not inside YouTube, not a browser extension, not a tab.

It is not technically possible	There is no surface inside YouTube to build on. An extension could inject interface elements, but it would not hold the data — watch history arrives via an export tied to the *child's* account, while the guardian is signed in as themselves.
It would undermine the product	Ember's argument is that the answer is not more YouTube. A product that lives inside YouTube contradicts its own thesis on sight.
It is the wrong moment	Guardians think about their child's week on their own phone, in a quiet moment. Not in the application their child is absorbed in.

> **THE CORRECT MENTAL MODEL**
>
> Ember is not a dashboard. A dashboard is somewhere you go to check on things. Ember is something that **arrives**. The weekly experience is: notification on Thursday, open it, read one card, print, done — approximately ninety seconds, once a week. The Curiosity Graph is not the home screen; it is what a guardian looks at once after onboarding and occasionally thereafter. Promoting it to the landing page converts the product back into an analytics tool.

9.2 Complete screen flow

FIRST RUN — HAPPENS ONCE



1 · Sign in
Name the household

2 · Add child
Name, age, privacy note

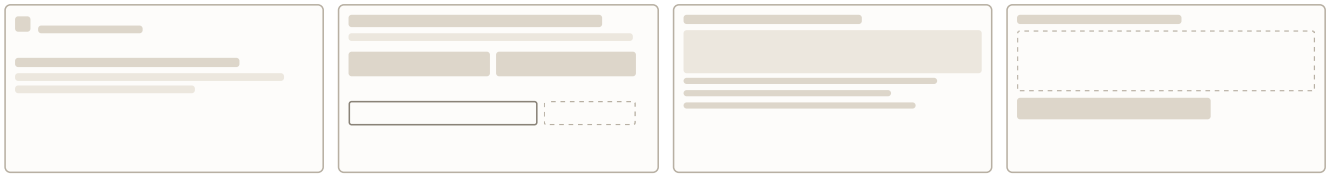
3 · Choose depth
One tap, export, or save

4 · Handoff
Leaves to Google, place held

5 · Drop file
Confirm it is the right child

6 · Understanding
Live list, then first reveal

EVERY WEEK — THE WHOLE PRODUCT, IN NINETY SECONDS



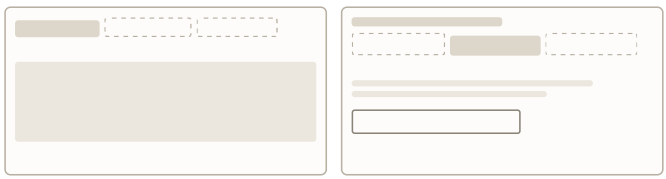
7 · Notification
Thursday. One per week

8 · This week
Home. One card, one decision

9 · Quest Card
Print view. Leaves the screen

10 · Log it
One tap, photo optional

ANYTIME — NOBODY IS REQUIRED TO GO HERE



11 · Curiosity
Per child, or both

12 · Report
Pick audience, export PDF

Twelve screens in total. Six are one-time setup; four carry the entire recurring product; two are optional depth reached only from the foot of screen 8.

Screens 7 to 10 **are** Ember. Built against fixed data they would still demonstrate the proposition, which makes them the schedule's safety net.

9.3 Screen specification

#	SCREEN	PRIORITY	NOTES
1	**Sign in / household**	P0	Guardian authentication, household creation. Minimal.
2	**Add child**	P0	Name, age band, consent record. Must state the household boundary in plain words — it is part of the argument, not chrome.
3	**Choose depth**	P0	Three tiers presented plainly with their real cost: one tap, a file export, or just start saving. Default selection is tier 1.
4	**Guided handoff**	P1	Four numbered steps, deep link into the YouTube-specific Takeout page. Must state that the place is held, since the session will break.
5	**Drop file + confirm**	P0	Accepts the whole .zip — never ask a guardian to unzip and locate a file. Then shows top channels and asks whether this is the right child. **Cheapest insurance in the application.**
6	**Understanding**	P0	A four-minute wait. A spinner here kills the demonstration; a live list of videos being understood turns dead time into the most persuasive moment in the product.
7	**Notification**	P1	One per week. Thursday evening.
8	**This week**	**P0 — home**	The landing screen. One activity, the reasoning, ownership split by child, print and decline. Build this before the graph.
9	**Quest Card**	P0	Print stylesheet, single page, Imagen illustration.
10	**Log it**	P1	One tap, optional photo. Feeds the graph.
11	**Curiosity Graph + Common Ground**	**P0 — hero**	On camera for roughly fifty seconds of the demonstration video. Must be started in week 0. A rushed visualisation sinks the centrepiece beat. The child switcher carries a “both” tab, which is where Common Ground lives.
12	**Family Report**	P1	Audience selector, section redaction, PDF generation.
—	**Household settings**	P2	Members, consent revocation, deletion. Not in the flow above; reachable from the menu.

9.4 What the guardian screen deliberately omits

There is no row of metric cards. No hours-watched figure, no watch-time chart, no top-channels leaderboard. The moment those appear, a guardian reads the product as a surveillance report and Ember becomes a parental control application. The single quantity shown is *videos understood*, not hours watched — framing the system as comprehending rather than counting.

On screen 8, the decline control carries equal weight to the primary action. A product that assumes compliance manufactures guilt, and guilt is why parenting applications get deleted. Refusal without friction is what makes the weekly card read as a suggestion rather than an assignment.

9.5 Design principles

- **Warm, not clinical.** The reference point is a well-made printed report, not a business intelligence dashboard. Off-white surfaces, one accent, generous whitespace, no charts.
- **Reasoning is always one tap away.** Every generated output expands to show why. Trust is the product.
- **One decision per week.** Resist a second call to action anywhere on screen 8.
- **Depth is opt-in.** A guardian who never leaves screen 8 still receives the full value.
- **The child's artifact is paper.** Hold the line.

9.6 Implementation notes

Build order	Screen 8 before screen 11. It is simpler, it is what a real guardian sees ninety-five per cent of the time, and completing it early guarantees something demonstrable even if the graph visualisation runs late.
Graph rendering	Do not use a live physics simulation. A force layout looks impressive until a node lands badly on the take being filmed. Compute the layout once, seed it deterministically, and animate nodes into fixed positions over roughly 800 ms — the same sense of computation, with a result that can be rehearsed.
Responsive split	Mobile-first, since that is where the weekly moment genuinely occurs. For filming, shoot the notification and weekly card in a phone frame and the Curiosity Graph on desktop where it has room. Same application, two contexts.
Navigation	Four items. This week · Curiosity · Reports · Household. “This week” is home.

> **THE DEMONSTRATION PATH IS NOT THE USER PATH**
>
> In the video the route is 2 → 3 → 5 → 6, then directly to 11 for the graph reveal and the Common Ground moment, then 9 for the printed card. Screen 4 is skipped because it is a handoff to a third-party website; screens 7, 8 and 10 are skipped because they are the ordinary weekly loop rather than the story. Put this path on a whiteboard before filming so nobody builds a screen the video never visits.

10 Judging Criteria Alignment

CRITERION	WEIGHT	HOW EMBER ADDRESSES IT
Technical Merit & Gen AI Implementation	**40%**	**Meaningful use:** native multimodal video understanding is the foundation, not a wrapper — remove Gemini and there is no product. Four distinct generative roles: comprehension, synthesis, grounded planning, safety adjudication, plus image generation for the artifact. **Implementation:** asynchronous agentic pipeline on ADK, deployed to Cloud Run, Firestore-tenanted, Search-grounded. **Scalability:** the shared video cache means marginal cost per household falls as the corpus grows. Per-family tenancy is the natural sharding boundary.
Problem Alignment & Impact	**25%**	Directly on theme — transforms how content is *understood* and *experienced*. Impact is measurable at the household level: activities proposed, activities completed, hours redirected. The beneficiary is a child's development, not an engagement metric.
Innovation & Creativity	**25%**	**Differentiation:** every competing product in this space measures or restricts. Ember redirects. **Innovative Gen AI:** Common Ground — semantic intersection of two children's inferred interest models to synthesise a shared activity — has no equivalent we are aware of, and is only reachable through theme-level rather than tag-level understanding.
User Experience & Solution Design	**10%**	One decision per week for a time-poor guardian. Reasoning always inspectable. The design constraint that every AI output terminates offscreen produces coherence between the stated problem and the delivered experience.

Mandatory submission requirements

Google AI models	Gemini Flash and Pro, Imagen, and Vertex AI Embeddings — all via Vertex AI
Agentic platform	Agent Development Kit, four-agent graph
Deployment	Cloud Run (services) and Firebase Hosting (client)
Proposal deck	PDF, owned by Design/Pitch from day one
Functional prototype	Deployed, publicly reachable
Demo video	3 minutes, public link
Language	All materials in English

11 Scalability & Sustainability

Unit economics improve with scale

Video analysis is the dominant cost and is cached globally by video ID. Children's viewing is heavily concentrated in a comparatively small set of popular content, so cache hit rate climbs steeply with adoption. The thousandth household costs materially less to serve than the first. This is a genuine structural advantage and should be stated in the pitch.

Technical scaling path

- Cloud Run scales horizontally with no state held in the container
- Cloud Tasks absorbs ingestion bursts; a household onboarding is naturally batched
- Firestore tenancy by `familyId` is the natural sharding boundary
- Flash for volume comprehension, Pro reserved for low-frequency synthesis — cost scales sub-linearly with history size

Commercial model

Household subscription, directly analogous to a family streaming plan and instantly legible to any evaluator. Alignment is clean: the customer is the guardian, the product's success metric is activities completed offline, and there is no advertising incentive to increase consumption. Adjacent revenue — partnerships with local class providers surfaced through Quest Planner — is deliberately deferred, because it introduces a commercial pressure the Guardian agent currently guarantees against.

Roadmap beyond the prototype

Near term	Commingled-history viewer clustering; additional platforms; multi-language Quest Cards
Medium term	Longitudinal skill ladders across months; school-holiday planning; accessibility adaptations for neurodivergent children
Long term	Anonymised, opt-in aggregate research on the relationship between content diet and offline activity — strictly separate from the product data path

12 Delivery Plan

26 calendar days to 19 October. Deck, video and documentation realistically consume five of them, so the true build window is approximately three weeks.

12.1 Team allocation

MEMBER	OWNS	DELIVERABLES
Backend / GCP	Ingestion and orchestration	GCP project and IAM, Takeout parser, Data Portability OAuth flow, Cloud Run service, Cloud Tasks fan-out, Firestore schema and rules, Cloud Scheduler
ML / Data	Content Sense and Curiosity Graph	Video-understanding prompt and locked output schema, caching strategy, theme synthesis, embeddings, Common Ground intersection logic
Frontend	Guardian experience	Firebase Hosting SPA, graph visualisation, Common Ground view, Quest Card print renderer, processing state, report builder
Design / Pitch	Narrative and evidence	Synthetic personas and histories, deck, demo script, 3-minute video, documentation. **Begins day one.**

12.2 Schedule

WINDOW	FOCUS	GATE
24–25 Sep	**Feasibility spike**	**Gate 0.** Twenty real videos through Gemini with one prompt. Do the themes come back non-obvious? If they read as tags, the concept is adjusted now rather than discovered in week three. *Nothing else is built until this passes.*
26–28 Sep	Foundations	Synthetic personas and both watch histories written and signed off. Output schema locked. GCP project live. Wireframes done.
29 Sep – 5 Oct	Thin vertical slice	**Gate 1.** One history in, one Quest Card out, end to end, deployed on Cloud Run. Visually ugly is acceptable. Slipping this gate is a schedule emergency.
6–12 Oct	Hero features	**Gate 2.** Feature complete: Common Ground, Search grounding, Guardian agent, report export, graph visualisation finished.
13–14 Oct	Harden	**Code freeze 14 Oct.** Bug fixes only thereafter.
15 Oct	Rehearse	Demo runs cold five consecutive times without intervention. Gemini cache pre-warmed for every demo video.
16 Oct	Film	A full day. Filming always takes one.
17–18 Oct	Edit and submit	**Submit 18 October.** Never on deadline day — portals fail.

12.3 Synthetic data strategy

The two synthetic watch histories are the most important design artifact in the project and are a writing task, not an engineering one. The sibling intersection must land in a narrow band: too obvious and the demonstration is flat, too abstract and it reads as hallucination.

Method	Write the two children as characters first, then work backwards to the histories that would produce them.
Worked shape	A ten-year-old immersed in survival and shelter-building; an eight-year-old on baking shorts and slime chemistry. Unrelated on the surface. The graph converges on *making things work with limited ingredients* and proposes campfire cooking, in which both children own a component.
Critical rule	**Fake histories, real videos.** Every record must point at a genuine YouTube URL that Gemini actually processes. The watch log is invented; the content understanding is real. State this plainly in the video — an honest synthetic setup earns more credit than a suspiciously flawless live demonstration.
Realism	Include repeats, Shorts, a few junk videos and one abandoned rabbit hole. Clean data looks manufactured.

12.4 Cut list, in order

If the schedule slips, sacrifice in this sequence: local-events Search grounding → audience re-framing on reports → more than two children → Imagen illustration → any authentication beyond a single hardcoded household.

Never cut: real video understanding, the Curiosity Graph reveal, Common Ground, the printed Quest Card. These four are the demonstration.

13 Demonstration Plan

13.1 Three-minute video, beat by beat

TIME	BEAT	CONTENT
0:00–0:20	The problem	A parent states it. Hours watched, nothing learned, no idea what to do about it.
0:20–0:35	Connect	Household created, two children added, consent boundary visible on screen.
0:35–1:00	Understanding	Gemini reading actual videos. Show the structured extraction appearing. Not a spinner.
1:00–1:50	**The reveal**	The graph builds. Peak on the unspoken-interest statement — the thing the parent did not know. The visualisation is the evidence; the sentence is the moment.
1:50–2:25	**Common Ground**	Second graph appears, themes intersect, shared activity generated. Emotional payoff.
2:25–2:40	Offscreen	Quest Card printing. Real hands, real paper.
2:40–3:00	Close	Loop closing, privacy model, scale story.

13.2 Pitch order differs from video order

Live, there is a narrator, question handling and the ability to build tension — so lead with the Curiosity Graph and treat Common Ground as the "and here is what that unlocks" reveal. In the video there is no narrator safety net, so the arc must keep moving. Same content, different pacing. The two do not need to match.

13.3 Prepared answers

"Why doesn't Google build this?"	It is YouTube recommending less YouTube. The incentive misalignment is structural, which is precisely why a third party operating on parent-authorised exported data is the right home for it.
"Isn't this surveillance?"	The guardian already holds export rights over a supervised child's account. Ember never shows a raw feed of what the child watched — it shows derived themes with reasoning. Outsiders get documents, never accounts.
"How do you get the data?"	Three tiers. Default is one OAuth tap on the child's account for liked videos and subscriptions. Deeper is full watch history via Takeout or the Data Portability API. Lightest is the guardian saving a video directly. The YouTube Data API cannot serve watch history — that playlist has returned an empty placeholder since 2016 — which is why the deeper tier goes through export rather than an API call.
"Isn't Takeout too hard for real families?"	Agreed, which is why it is not the front door. Tier 1 is a single sign-in with no file handling. Takeout is the option for households that want the richest signal.
"Why not just let parents save videos?"	Because then the product can only tell a parent what they already told it. Saves are declared intent; the graph exists to surface what nobody declared. Saving is an input, and on save Ember tells the guardian which child it fits and why.
"Is the demo data real?"	The histories are synthetic and authored. Every video in them is real and genuinely processed by Gemini. Stated up front, in the video.
"What if the AI suggests something unsafe?"	The Guardian agent is a mandatory terminal gate on every child-facing artifact, and the guardian approves or rejects every activity before a child ever sees it.

14 Risk Register

RISK	SEVERITY	MITIGATION
Themes come back generic — the graph is a tag cloud	**Critical**	The entire product rests on this assumption. Gate 0 on 24–25 September tests it directly before any infrastructure is written. Reasoning and cited evidence are enforced in the output schema, not requested in the prompt.
Sibling overlap is trivially obvious	High	Personas authored deliberately for surface divergence and theme convergence. Reviewed by the full team before histories are written.
Gemini video latency breaks the demo	High	Asynchronous from the first commit. Cache pre-warmed for all demonstration videos before filming.
Quest Planner hallucinates local options	High	Search grounding is mandatory, with grounded references persisted alongside the activity. If grounding is cut, the planner falls back to home-based activities requiring no external venue.
Read as "another parental control app"	Medium	Language discipline throughout. Redirect, never restrict. The offscreen constraint makes the distinction visible rather than merely claimed.
Graph visualisation rushed	Medium	It occupies roughly 50 seconds of the video. Started week 0, not week 2.
Time lost to authentication and tenancy	Medium	Schema modelled correctly, implementation kept minimal. Single-guardian path only.
Tier 1 blocked on supervised accounts	Medium	The likes and subscriptions surface is confirmed on the public API but not yet on a supervised child account. Validated this week against a standard test account; if supervised access is restricted, tier 2 becomes primary onboarding and the friction argument is made honestly rather than solved.
Likes signal too sparse	Medium	A child may like few videos or none. Subscriptions compensate, since channel choice is deliberate and a channel's recent uploads can be analysed to characterise it. If both are thin, Ember prompts the guardian to add tier 2.
Product drifts into a bookmarking tool	Medium	Guardian saves are seductive because they are frictionless, but a product built on them can only reflect what the guardian already knew. Saves are permanently scoped as an input to the graph, never as the source of truth.
Data Portability approval unavailable	Low	Takeout upload ships regardless and requires no approval. The API is demonstrated in testing mode against team accounts.
Ethics challenge from the panel	Low	Raised proactively by the team, with the consent chain and household boundary as the answer. Never allow an evaluator to raise it first.

15 Appendix A — Naming

The working name throughout this document is **Ember**. A video leaves a spark of interest; unattended, it goes out. Ember catches it. The metaphor also ties to the campfire-cooking hero demonstration.

CANDIDATE	RATIONALE	TRADE-OFF
Ember	Short, warm, memorable. The spark-that-fades metaphor maps exactly onto the problem. Pairs cleanly with feature names.	Fire metaphors are not rare. Requires the tagline to carry the specificity.
Offscreen	Punchiest and most memorable. Literally states the thesis.	Frames around absence and restriction, which fights the redirect-not-restrict position.
Kindling	Strongest metaphor of the set — a spark alone dies, kindling turns it into a fire.	Confusable with an established e-reader brand.
Hearth	Warmest. Directly evokes the family gathering place.	Says nothing about curiosity or discovery. Softest of the four.

Feature nomenclature

Curiosity Graph	The per-child inferred interest model. Keep regardless of product name — it is descriptive, memorable and does real work in the pitch.
Common Ground	The sibling intersection. Warm, accurate, immediately understood.
Quest Card	The printable child-facing artifact.
Family Report	The shareable PDF. Health-report framing.
Guardian Agent	The safety gate. Deliberate double meaning with the guardian role.

The product name appears in the title, cover, application shell and deck. Feature names are unaffected by a change, so a swap is a single pass.

Appendix B — Decision Log

DECISION	REASONING
Household as tenant, not individual	Enables Common Ground; matches how families actually operate; gives a legible subscription model.
Guardians only, no extended family	An aunt with her own children has no stake in this child's development. Privacy by scope.
Share by artifact, not by access	An account grants ongoing visibility; a document does not. Simpler to build and the stronger privacy position.
Video understanding, not metadata	Titles cannot distinguish instruction from entertainment. This is the foundation of everything above it.
AI proposes the activity, guardian edits	Handing synthesis back to the user offloads the hard part and signals the model does analysis but not reasoning.
Weekly, not monthly	A monthly report is too slow to build a habit and too slow to demonstrate. One weekend, one thing.
Four agents, one Cloud Run service	Preserves the agentic architecture while leaving one deployment and one log stream to debug under pressure.
Synthetic histories, real videos	Deterministic demonstration with genuine model work. Honest and reproducible.
Three input tiers, tier 1 as default	Takeout as the front door would lose most non-technical households. One OAuth tap for likes and subscriptions preserves the child-sourced signal while removing almost all of the friction.
Saves are an input, not the source	A product built on guardian saves can only reflect what the guardian already believed. The graph exists to surface what nobody declared.
Graph as primary, Common Ground as payoff	Not ranked features — setup and payoff in one continuous arc. The graph supplies credibility, the activity supplies resolution.